MIDI-ANALYSER-BLE

User Guide · Firmware v4.0

MINDBURNER MIDI PRODUCTS

Hand-built in Belfast, Northern Ireland

Welcome

The MIDI-ANALYSER-BLE decodes every MIDI message in real time on a 1.8" colour screen. Plug it into any 5-pin DIN MIDI source, or start it in Bluetooth mode to go wireless. No computer or software needed.

Getting Started

Power	9–12V DC centre-positive, or USB.
MIDI IN	Connect the MIDI OUT of your keyboard, sequencer or interface. MIDI THRU passes the signal on unchanged.
First boot	An animated logo plays, then the last-chosen Boot App opens (Analyser by default).

The three buttons

A	Main action for the current app. A **hold** (1 second) usually resets or exits.
B	Second action (pause, filter, change value, switch page).
C	Next app. A short banner shows its controls; press C again to skip it.
A + C hold 3s	Reboot straight into Bluetooth mode, from anywhere.

DIN or Bluetooth

The unit starts in DIN mode. To use Bluetooth, hold **C** while powering on (or use the A+C shortcut). Pair with **MB_ANALYSER**. A cold power-up returns to DIN. Bluetooth works with a CME WIDI BUD (not included) or any BLE MIDI host.

The Apps

1. Analyser

A scrolling list of decoded messages: type, note, values and channel, colour-coded, with velocity bars. The 16 squares at the top light when each channel is active.

A	Pause. Press again to step through events one at a time. Step view shows full detail, including the time since the previous event.
A hold	Back to live view.
B	Filter screen. A cycles message type. Press B to move to the **Channel** box, A cycles ALL / 1–16, B again to finish.

With a channel chosen, system messages (clock, Start/Stop, SysEx) are hidden, and that channel is outlined red in the activity strip. Clock and Active Sensing are hidden unless chosen in the filter.

2. Raw View

Four modes, cycled with A: **HEX** (last bytes, status bytes highlighted), **BINARY**, **DECODED** (last message, SysEx size, bytes per second and **bus load** as a % of DIN capacity) and **CAPTURE** (grabs one complete message).

B	Pause/resume HEX and BINARY.
A hold (HEX)	Scroll back a page; keep holding to keep paging.
A hold (CAPTURE)	Re-arm for the next message.

Device ID: if CAPTURE grabs a SysEx Identity Reply it decodes the manufacturer, family, model and firmware version. The analyser has no MIDI out, so send the request `F0 7E 7F 06 01 F7` to the device from a computer and route the device's MIDI OUT into the analyser. Not every device replies.

3. Clock Meter

Turn WiFi off for best timing accuracy. Shows tempo in BPM, RUN/STOP state, four beat lights, a tempo history graph, and jitter (spread and standard deviation in microseconds).

A	Full-screen big BPM.
B	Groove Meter page.
A hold	Reset all clock and groove readings.

Groove Meter

Shows how far each Note On lands from the nearest 16th of the incoming clock: average offset (+ late, – early), spread, separate even/odd 16th averages, a **swing %** (50 = straight, MPC style) and an early/late histogram.

It needs notes and clock on the same input, and a known position: send Start (or Song Position + Continue). The header shows LOCKED when ready, NO POS if no position is known. Accuracy is about ±0.5 ms.

4. MIDI Monitor

A 16-channel dashboard: last note (green while held), volume bar (CC 7) and pitch bend direction. A cell turns red if a note has been held over 8 seconds with no Note Off: a likely stuck note.

5. Note Histogram

How often each of the 12 notes has been played. Useful for spotting key or scale. A hold resets.

6. CC Scope

A live graph of one controller. A steps through 23 common CCs and pitch bend.

7. CC Learn

Move any knob, fader or pedal and it locks on, showing CC number, channel, value, a name if recognised, and a live graph. A re-arms, B pauses, A hold clears.

8. Programs

Last Program Change on each channel (shown 1–128) with bank MSB. A cell flashes yellow when a new one arrives. A selects a channel; the footer then shows full bank MSB:LSB, program and raw value. A hold clears. Handy for checking patches load correctly at the start of a gig.

9. MIDI Health

Orphan data	Data bytes with no status. A couple at power-up are normal if plugged in mid-stream.
Interrupted	Messages cut short by a new one.
SysEx unterm	SysEx dumps that never finished properly.
Undefined st	Status bytes the MIDI spec doesn't define.
UART frm/brk	Electrical/framing errors: suspect cables or connectors.
UART overflow	Data arrived faster than it could be read.
Link lost	Active Sensing dropouts (see below).
Running st	Messages using running status. Information only, not an error.
Bus load	% of DIN capacity used, plus peak.

Error counts show green at 0 and red otherwise. The coloured bar at the bottom shows what filled the cable over the last second: notes, CC, other channel messages, clock, Active Sensing, SysEx. A hold resets everything.

LINK LOST: if a device has been sending Active Sensing and it stops for over 300 ms, LINK LOST flashes in every app until it returns. Gear that never sends Active Sensing never triggers it (header shows AS:--). If you swap devices on purpose, hold A on this page to clear it.

10. Settings

Brightness	20–100%.
WiFi Monitor	On/off. Applies after reboot.

Boot App	Which app opens at power-up.
Auto-Dim	Off / 30s / 1 min / 5 min with no button presses or note/CC traffic. Clock and Active Sensing alone don't keep it awake.
Reboot	Hold A on this row.

A selects a row, B changes it. Settings save automatically.

WiFi Live Monitor

Turn on in Settings and reboot. Join network **MB_ANALYSER_MON** (password **midi1234**) and browse to **192.168.4.1**. You get a colour-coded live log with hex, Pause, Clear, Auto-scroll and Show Clock, a status line with source, BPM and bus load, and **Download CSV** for the last 256 events. Turn WiFi off when measuring clock timing.

Firmware Updates

Free updates are installed over USB from the Mindburner web page in a Chrome or Edge browser. No software to install.

Troubleshooting

Nothing on screen	Check MIDI OUT → MIDI IN, and the right MIDI channel/port on the sending device.
Clock shows STOP	No Start received, or the clock stopped for over 2 seconds.
Groove says NO POS	Press Start (or send Song Position + Continue).
LINK LOST won't clear	Hold A on MIDI Health.
UART errors climbing	Try another cable, or check the sender's connector.
Bluetooth won't pair	Reboot with A+C held 3s, then pair to MB_ANALYSER.